DISCRETE STRUCTURES

CMSC250

Chapter 9: Fundamental Proof Techniques

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Direct proof, contraposition, contradiction, and equivalence



Contents

1 What a complete proof looks like **3**

2 Direct proofs: start with the hypothesis **3**

3 Contraposition: negate in the correct order **4**

4 Contradiction: what BWOC actually means **5**

5 Equivalence proofs: two directions **6**

6 Counterexamples: when a universal claim is false **6**

7 Proof templates you can reuse **7**

8 Practice problems **7**

9 Detailed practice solutions **7**

1 What a complete proof looks like

A proof is not simply a calculation ending at the right expression. It is an argument establishing a claim from definitions and previously accepted results. A well-written proof follows a recognizable structure: choose arbitrary objects, state the assumption, announce what must be shown, apply definitions, cite closure, explain the conclusion, and finish with a tombstone ■.

1. **Declare objects and domains:** "Let $a, b \in \mathbb{Z}$, selected arbitrarily."
2. **For an implication, assume its left-hand side:** "Assume $a \mid b$."
3. **Identify the target:** "I will show $a^2 \mid b^2$."
4. **Unpack the definition:** "There is $k \in \mathbb{Z}$ with $b = ak$."
5. **Derive the required form:** $b^2 = a^2k^2$.
6. **Check domain closure:** $k^2 \in \mathbb{Z}$.
7. **Conclude using the definition, generalize, and place ■.**

Key idea. The tombstone ■ marks the end of a completed proof, not the point at which you stop explaining. Always justify the conclusion first.

2 Direct proofs: start with the hypothesis

For $P \to Q$, assume P and derive Q. Never assume Q and work backward without explaining why the steps are reversible.

Worked example: divisibility survives squaring

Claim. For all $a, b \in \mathbb{N}$, if $a \mid b$ then $a^2 \mid b^2$.

Proof. Let $a, b \in \mathbb{N}$, selected arbitrarily. Assume $a \mid b$. [I will show $a^2 \mid b^2$.] By the definition of divisibility, $b = ak$ for some $k \in \mathbb{N}$. Therefore

$$b^2 = (ak)^2 = a^2k^2.$$

Note that $k^2 \in \mathbb{N}$ by closure under multiplication. Thus b^2 is a multiple of a^2, so $a^2 \mid b^2$ by the definition of divisibility. Since a, b were arbitrary, the proposition holds for all such natural numbers. ■

Why this works. The essential step is not merely squaring: the witness k^2 must belong to the required number system.

Worked example: divisibility of a difference

Claim. For all integers a, b, c, if $a \mid b$ and $a \mid c$, then $a \mid (b - c)$.

Proof. Let $a, b, c \in \mathbb{Z}$ be arbitrary. Assume $a \mid b$ and $a \mid c$. [I will show $a \mid (b - c)$.] By divisibility, $b = ak$ and $c = am$ for some $k, m \in \mathbb{Z}$. Then

$$b - c = ak - am = a(k - m).$$

Since $\mathbb{Z}$ is closed under subtraction, $k - m \in \mathbb{Z}$. Thus $a \mid (b - c)$ by the definition of divisibility. The variables were arbitrary, so the claim follows. ■

Worked example: a universal parity proof

Claim. For every odd natural number n, the number $n + 11$ is even.

Proof. Let a be an odd natural number, selected arbitrarily. [I will show $a + 11$ is even.] Since a is odd, $a = 2k + 1$ for some $k \in \mathbb{N}$. Therefore

$$a + 11 = (2k + 1) + 11 = 2(k + 6).$$

Since $k + 6 \in \mathbb{N}$ by closure under addition, $a + 11$ is twice a natural number and hence even. Since a was arbitrary, the claim holds for every odd natural number. ■

Why this works. The opening and closing sentences are the mechanism of universal generalization.

Worked example: even input, odd output

Claim. For every even integer n, $n^3 + 5$ is odd.

Proof. Let $n \in \mathbb{Z}$ be even, selected arbitrarily. Then $n = 2k$ for some $k \in \mathbb{Z}$. Compute

$$n^3 + 5 = (2k)^3 + 5 = 8k^3 + 5 = 2(4k^3 + 2) + 1.$$

Because $4k^3 + 2 \in \mathbb{Z}$ by closure under addition and multiplication, the displayed expression has the form $2m + 1$ with $m \in \mathbb{Z}$. Thus $n^3 + 5$ is odd. Since n was arbitrary, the claim follows. ■

3 Contraposition: negate in the correct order

The contrapositive of $P \to Q$ is $\sim Q \to \sim P$. It is logically equivalent to the original implication. If Q is a disjunction, De Morgan's law turns its negation into a conjunction: $\sim (A \vee B) \equiv (\sim A \wedge \sim B)$.

Worked example: if a product is even, a factor is even

Claim. For all integers a, b, if ab is even, then a is even or b is even.

Proof. Let $a, b \in \mathbb{Z}$ be arbitrary. We prove the contrapositive. Assume a is odd *and* b is odd. [I will show ab is odd.] Write $a = 2k + 1$ and $b = 2m + 1$ for some $k, m \in \mathbb{Z}$. Then

$$\begin{aligned} ab &= (2k + 1)(2m + 1) \\ &= 4km + 2k + 2m + 1 \\ &= 2(2km + k + m) + 1. \end{aligned}$$

By closure, $2km + k + m \in \mathbb{Z}$. Hence ab is odd. This establishes the contrapositive and therefore the original implication. Since a, b were arbitrary, the claim holds universally. ■

Why this works. Negating "a is even or b is even" gives "a is odd and b is odd." Replacing *and* by *or* would change the proof.

Worked example: An additional contrapositive proof

Claim. For any integer n, if n^2 is even, then n is even.

Proof. Let $n \in \mathbb{Z}$ be arbitrary. Prove the contrapositive: assume n is odd. Then $n = 2k + 1$ for an integer k, and

$$n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

The quantity $2k^2 + 2k$ is an integer by closure, so n^2 is odd. Thus the contrapositive is true, establishing the original implication. ■

4 Contradiction: what BWOC actually means

To prove S by contradiction, suppose (BWOC) that $\sim S$ holds, and derive an impossibility. For an implication, first assume the hypothesis, then suppose the conclusion is false. The contradiction must be explicitly named.

Worked example: an irrational fifth root remains irrational

Claim. For every positive irrational real number x, $\sqrt[5]{x}$ is irrational.

Proof. Let x be a positive irrational number, selected arbitrarily. Suppose (BWOC) that $\sqrt[5]{x} \in \mathbb{Q}$. By the definition of rational, $\sqrt[5]{x} = a/b$ for some integers a, b with $b \neq 0$. Raising both sides to the fifth power gives

$$x = \frac{a^5}{b^5}.$$

The numerator a^5 and denominator b^5 are integers by closure of $\mathbb{Z}$ under multiplication. Furthermore $b^5 \neq 0$ since $b \neq 0$. Thus x is a quotient of two integers with nonzero denominator, so $x \in \mathbb{Q}$. This contradicts the assumption that x is irrational. Hence $\sqrt[5]{x}$ is irrational. Since x was arbitrary, the claim follows. ■

Why this works. Notice the two separate checks: (1) numerator and denominator are integers; (2) denominator is nonzero. Both are essential to invoking the definition of rational.

Worked example: a divisibility contradiction without modular arithmetic

Claim. For every integer m, if $7 \mid m$, then $7 \nmid (m+6)$.

Proof. Let $m \in \mathbb{Z}$ be arbitrary and assume $7 \mid m$. [I will show $7 \nmid (m+6)$.] Suppose (BWOC) that $7 \mid (m+6)$. Then there are integers k, t with $m = 7k$ and $m + 6 = 7t$. Subtracting gives

$$6 = (m+6) - m = 7t - 7k = 7(t-k).$$

Since $t - k \in \mathbb{Z}$, this implies $7 \mid 6$, which is impossible: no integer multiple of 7 equals 6. This is a contradiction, so $7 \nmid (m+6)$. Since m was arbitrary, the implication holds for every integer. ■

Worked example: A contradiction using a rational expression

Claim. If x is irrational, then $(3x+7)/2$ is irrational.

Proof. Let $x \in \mathbb{R}$ be irrational. Suppose (BWOC) that $(3x+7)/2$ is rational. Then $(3x+7)/2 = a/b$ for integers a, b with $b \neq 0$. Rearranging,

$$x = \frac{2a - 7b}{3b}.$$

Both $2a - 7b$ and $3b$ are integers by closure. Also $3b \neq 0$ since $b \neq 0$. Therefore x is rational by definition, contradicting the assumption. Thus $(3x+7)/2$ is irrational. ■

Key idea. Grading detail: saying $b \neq 0$ at the beginning does not automatically explain why a *new* denominator such as $3b$ is nonzero. State that reasoning explicitly.

5 Equivalence proofs: two directions

To prove $P \leftrightarrow Q$, prove both $P \to Q$ and $Q \to P$. You may use contraposition for either direction, but identify which direction is being proved.

Worked example: parity of neighboring integers

Claim. For every integer n, n is odd if and only if $n+1$ is even.

Proof. Let $n \in \mathbb{Z}$ be arbitrary. **Forward:** Suppose n is odd. Then $n = 2k+1$ for an integer k, so $n + 1 = 2(k+1)$ is even, since $k + 1 \in \mathbb{Z}$. **Reverse:** Suppose $n + 1$ is even. Then $n + 1 = 2m$ for an integer m, hence $n = 2(m-1) + 1$ is odd, since $m - 1 \in \mathbb{Z}$. Both directions hold, so the biconditional is true. ■

6 Counterexamples: when a universal claim is false

One counterexample disproves a universal claim. It must satisfy *all hypotheses* and violate the conclusion.

Worked example: two divisors need not multiply to a divisor

The claim "If $a \mid c$ and $b \mid c$, then $ab \mid c$" is false. Choose $a = b = c = 2$. Both $2 \mid 2$ hypotheses hold, but $ab = 4$ and $4 \nmid 2$. Thus the claim is false. ■

Worked example: divisibility of a sum does not force a divisor of a summand

The claim "If $a \mid (b + c)$, then $a \mid b$ or $a \mid c$" is false. Take $a = 3$, $b = 1$, $c = 2$. Then $3 \mid (1 + 2)$ but $3 \nmid 1$ and $3 \nmid 2$. ■

7 Proof templates you can reuse

Direct implication: Let variables be arbitrary. Assume P. [I will show Q.] Apply definitions and algebra. Cite closure. Therefore Q. Since variables were arbitrary, the claim holds. ■

Contrapositive: Let variables be arbitrary. I prove the contrapositive. Assume $\sim Q$. [I will show $\sim P$.] Derive $\sim P$. Therefore the original implication holds. ■

Contradiction: Assume the hypotheses. Suppose (BWOC) the desired conclusion is false. Derive a statement contradicting an assumption or established fact. Therefore the supposition is false and the desired conclusion holds. ■

8 Practice problems

1. Prove directly: if $5 \mid n$, then $5 \mid (3n + 10)$.
2. Prove directly: if $a \mid b$ and $b \mid c$, then $a \mid c$.
3. Prove by contraposition: if n^2 is odd, then n is odd.
4. Prove by contradiction: if $5 \mid m$, then $5 \nmid (m + 2)$.
5. Prove directly that the sum of two odd integers is even.
6. Give a counterexample: if $a \mid bc$, then $a \mid b$ or $a \mid c$.
7. Prove that n is even if and only if $n + 3$ is odd.
8. Explain the logical mistake in negating "a is odd and b is odd" as "a is even and b is even."

9 Detailed practice solutions

1. Let $n \in \mathbb{Z}$ be arbitrary and assume $5 \mid n$. Write $n = 5k$ for $k \in \mathbb{Z}$. Then $3n + 10 = 5(3k + 2)$; closure gives $3k + 2 \in \mathbb{Z}$. Thus $5 \mid (3n + 10)$. ■
2. Let $a, b, c \in \mathbb{Z}$ be arbitrary. Assume $a \mid b$ and $b \mid c$. Write $b = ak$, $c = bm$ for integers k, m. Then $c = a(km)$ and $km \in \mathbb{Z}$, hence $a \mid c$. ■
3. Contrapositive: if n is even then n^2 is even. For $n = 2k$, $n^2 = 2(2k^2)$, and $2k^2 \in \mathbb{Z}$. Therefore the contrapositive and original implication hold. ■

4. Let $m \in \mathbb{Z}$ and assume $5 \mid m$. Suppose (BWOC) $5 \mid (m+2)$. Then $m = 5k$, $m+2 = 5t$ for integers k, t, so $2 = 5(t-k)$, impossible. Thus $5 \nmid (m+2)$. ■

5. Write odd integers as $2k+1$ and $2m+1$. Their sum is $2(k+m+1)$; the parenthesized quantity is an integer by closure, hence the sum is even. ■

6. Choose $a = 6$, $b = 2$, $c = 3$. Then $6 \mid 6$ but $6 \nmid 2$ and $6 \nmid 3$. ■

7. Forward: $n = 2k$ gives $n+3 = 2(k+1)+1$, odd. Reverse: $n+3 = 2m+1$ gives $n = 2(m-1)$, even. Integer closure justifies both witnesses. ■

8. De Morgan's law gives $\sim (A \wedge B) \equiv \sim A \vee \sim B$. The correct negation is "a is even *or* b is even."